EMIPredict AI – Model Performance Report

Classification and regression model comparison, tracked with MLflow

Live app: https://abhisheknimaje435-emipredict-ai-app-ccio5a.streamlit.app

Repository: https://github.com/AbhishekNimaje435/emipredict-ai

1. Summary

Eight models were trained and logged in MLflow: four classifiers for EMI eligibility (Eligible / High_Risk / Not_Eligible) and four regressors for maximum monthly EMI. Model selection was done on the validation set; only the selected models were evaluated on the held-out test set, which keeps the test score an unbiased estimate.

XGBoost won both tasks and is the model registered and deployed in the Streamlit application.

Task	Selected model	Key result (test set)	Project target	Status
Classification	XGBoost Classifier	Accuracy 97.54%, ROC-AUC 0.997	Accuracy > 90%	Met
Regression	XGBoost Regressor	RMSE ₹746, R^2 0.9906	RMSE < ₹2,000	Met

2. Classification: EMI eligibility

2.1 Validation comparison

Model	Accuracy	F1 (macro)	F1 (weighted)	Precision (macro)	Recall (macro)	ROC-AUC	Train time (s)
XGBoost (selected)	0.9752	0.9119	0.9767	0.8855	0.9493	0.9969	166.7
Decision Tree	0.9403	0.8276	0.9486	0.7951	0.9087	0.9681	18.7
Random Forest	0.9209	0.7932	0.9350	0.7695	0.8919	0.9859	117.1
Logistic Regression	0.8601	0.7140	0.8910	0.7071	0.8394	0.9654	39.3

Class imbalance (77.3% Not_Eligible, 18.4% Eligible, 4.3% High_Risk) was handled with balanced sample weights, and models were ranked by macro F1 rather than accuracy.

2.2 Per-class recall (validation)

Model	Eligible	High_Risk	Not_Eligible
XGBoost	0.9597	0.9055	0.9828
Decision Tree	0.9054	0.8681	0.9527
Random Forest	0.8688	0.8708	0.9360
Logistic Regression	0.8021	0.8411	0.8749

2.3 Test-set result for the selected model (XGBoost)

Metric	Validation	Test
Accuracy	0.9752	0.9754
F1 (macro)	0.9119	0.9127
F1 (weighted)	0.9767	0.9768
Precision (macro)	0.8855	0.8860
Recall (macro)	0.9493	0.9499

Metric	Validation	Test
High_Risk recall	0.9055	0.9047
ROC-AUC	0.9969	0.9970

Validation and test scores are almost identical, so the selected model generalises well to unseen data.

2.4 Confusion matrix (XGBoost, validation set, 60,004 records)

Actual \ Predicted	Eligible	High_Risk	Not_Eligible
Eligible	10,588	385	60
High_Risk	128	2,348	117
Not_Eligible	69	727	45,582

58,518 of 60,004 records are classified correctly (97.5%). Only 60 Eligible applicants were predicted Not_Eligible and only 69 Not_Eligible applicants were predicted Eligible. The largest error is 727 Not_Eligible applicants predicted High_Risk, which lowers High_Risk precision to about 68% (2,348 of 3,460 predictions).

2.5 Feature importance (XGBoost classifier, approximate values from the top-15 chart)

Feature	Importance
emi_to_disposable	0.16
existing_loans (Yes)	0.09
existing_loans (No)	0.09
dti_x_credit	0.07
emi_to_income	0.05
affordability_ratio	0.03
monthly_salary (raw)	0.02

The engineered affordability ratios dominate the classifier. The two existing_loans columns come from one-hot encoding of one variable, so together they carry about 0.18, roughly the same weight as the top ratio.

2.6 Observations

- Logistic Regression is the interpretable baseline. Its 86% accuracy shows the eligibility boundary is non-linear (interactions between income, obligations and credit score), so linear models fall short.
- Decision Tree is surprisingly strong for its cost (18.7 s) but is a single tree and less stable.
- Random Forest scored below the Decision Tree on accuracy and macro F1 yet above it on ROC-AUC. This usually points to constrained hyperparameters (for example shallow depth or few trees) rather than an inherent weakness; further tuning could narrow the gap.
- High_Risk is the hardest class for every model. Macro precision (0.886) is far below weighted precision (about 0.98), which indicates that the smaller, borderline class is where most errors occur. XGBoost still reaches about 90.5% recall on it.

3. Regression: maximum monthly EMI

3.1 Validation comparison

Model	RMSE (₹)	MAE (₹)	R^2	MAPE (%)	Train time (s)
XGBoost (selected)	704.2	239.5	0.9918	7.17	46.9
Random Forest	1,071.8	320.8	0.9809	8.42	1,210.6
Decision Tree	1,312.1	447.6	0.9714	10.10	21.4
Linear Regression	3,556.4	2,428.1	0.7901	161.64	1.4

3.2 Test-set result for the selected model (XGBoost)

Metric	Validation	Test
RMSE (₹)	704.2	746.2
MAE (₹)	239.5	240.5

Metric	Validation	Test
R^2	0.9918	0.9906
MAPE (%)	7.17	7.26

Test RMSE is about 6% higher than validation, and MAE is virtually unchanged, so the model generalises well.

3.3 Feature importance (XGBoost regressor, approximate)

existing_loans (Yes) has an importance of about 0.65 and disposable_income about 0.14; every other feature is below 0.03. The predicted maximum EMI therefore depends mostly on whether the applicant already has loans and how much income is left after expenses.

3.4 Observations

- Linear Regression fails the project target (RMSE ₹3,556 vs ₹2,000) and has a MAPE above 160%. The maximum-EMI target is clearly non-linear in the inputs.
- Random Forest meets the target but took about 20 minutes to train, roughly 26 times longer than XGBoost, for worse accuracy.
- Decision Tree meets the target at very low cost but has roughly 1.9 times the RMSE of XGBoost.
- MAE (₹240) is much smaller than RMSE (₹746), meaning most predictions are very close and a minority of large misses drive the RMSE.
- The target is heavily right-skewed: it runs from near zero up to about ₹90,000, with about a third of the records at very small values. Small target values inflate MAPE, which is why MAPE looks high relative to the other metrics.

4. Best model selection and justification

Selection rule: highest validation score on the primary metric (macro F1 for classification, RMSE for regression), then confirmed once on the test set.

Why XGBoost for both tasks:

1. Best on every accuracy-type metric in both tasks on validation, with consistent test results.

2. Largest gain on the hard class: it leads on High_Risk recall (0.906) and macro F1 (0.912), which matters most because borderline applicants are where lending decisions are riskiest.
3. Meets the project targets with margin: accuracy 97.5% against 90%, and test RMSE ₹746 against ₹2,000 (about 63% lower than the limit).
4. Reasonable training cost: the regressor trained in 47 s versus 1,211 s for Random Forest. The classifier was the slowest of the four classifiers (166.7 s versus 117.1 s for Random Forest and 18.7 s for the Decision Tree), but training is an offline step and the extra time buys a clear accuracy gain.
5. Good generalisation: validation and test metrics agree closely.

Trade-offs: XGBoost is less interpretable than Logistic or Linear Regression. For regulated lending this is mitigated with feature-importance analysis (and optionally SHAP) so every decision can be explained and documented.

5. MLflow tracking and model registry

All 8 runs were logged with parameters, validation/test metrics, training time and artifacts (confusion matrix and feature-importance plots for tree models). Two experiments were used: EMI_Eligibility_Classification and EMI_Amount_Regression. The winning run of each experiment is tagged selected_for_production and is the only one evaluated on the test set.

The best model of each task is registered in the MLflow Model Registry with the production alias and is loaded by the Streamlit application for real-time prediction:

Registered model	Version	Alias	Run ID
EMI_Eligibility_Classifier	1	production	bdf73abb49d44071a6372fc86d282839
EMI_Amount_Regressor	1	production	6076b8ae7e2b4bb3b022e4bec810fd46

To open the MLflow UI locally from the project folder that contains mlflow.db:

mlflow ui --backend-store-uri sqlite:///mlflow.db --workers 1

The same comparison is available in the Model Performance page of the Streamlit app.

6. Limitations and next steps

- Improve High_Risk precision through class weighting or threshold tuning.
- Run a hyperparameter search on Random Forest, which appears under-tuned.
- Add SHAP explanations to support compliance and loan-officer trust.
- Monitor for data drift after deployment and retrain periodically.

This is an educational project. Its predictions are not a substitute for real lending decisions.